

Week 2

1.1 Conceptual Data Modeling

How to represent the real world?

Representation depends on purpose.

- What part of the world should we capture for our application?
- How do we represent it?
- There are many different ways (creative ways)

Idea to Database Application

Step 0: Plan Application. Understand domain/users.

Step 1: Conceptual data modelling.

Step 2: Physical data modelling.

Step 3: Create database. Develop Application.

Why conceptual modeling first?

- Focus on high-level
- close to how we think of the real-world.
- Use natural expression
 - sketch diagrams
- Bridge to physical modelling
 - Easy mapping to various physical data models.

Real-World

↓
conceptual

↓
physical

1.2 Example Applications

Ex1: Academic World

Who: students, courses, professors

What: taking courses, teaching courses, advising.

Ex2: Friday Night

Who: drinkers, bars, bars

What: drinkers go bar, drink beer, selling beer.

1.3 ER Model Basics

Entity - Relationship Model

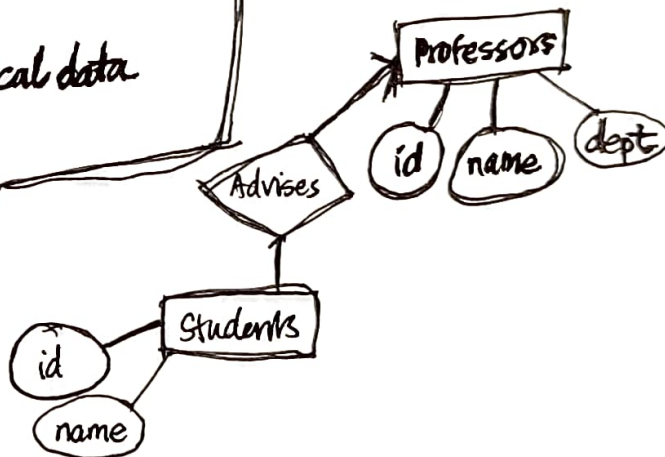
- create ER diagrams using entity sets, relationships and attributes.

Diagram language

- What information the database must hold.
- The components of that information.
- The relationships among the components of that information.

Output ER diagram.

Academic World



Entity sets and attributes

• Entity Sets

- Real-world objects distinguishable from others.
- Characterized by using a set of attributes.

• Attributes

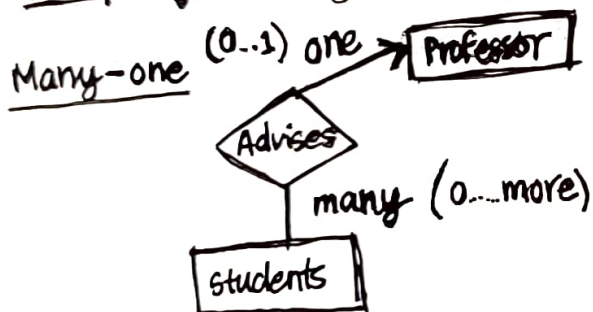
- Each attribute is a property of an entity
- Each attribute has a simple value: string, integers, reals etc.

Relationships

- Relates two or more entity sets for their interaction.
- Cardinality (or degree): Number of entity sets involved.
 - Binary: Two sets are related
 - Multiway: More than 2.
- A relationship can have attributes as its properties.

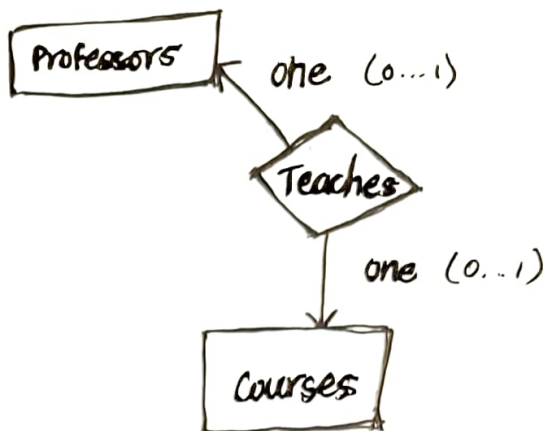
1.4 ER Model - More on Relationships

Multiplicity of Binary Relationships



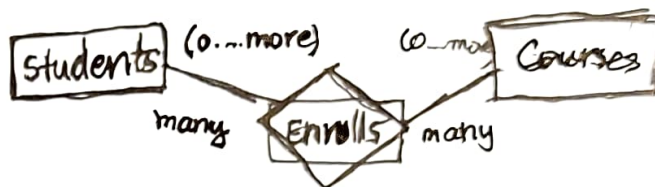
~~many~~ student has one advisory prof.
professor can advise many students

one-one



one professor teaches one course
one course is taught by one prof.

many-to-many

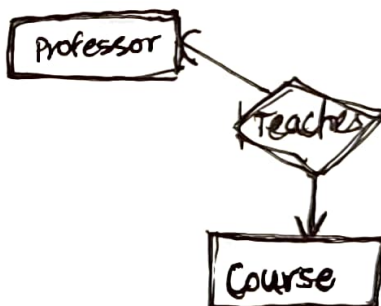


one student can be enrolled in many courses.

one course can have many students

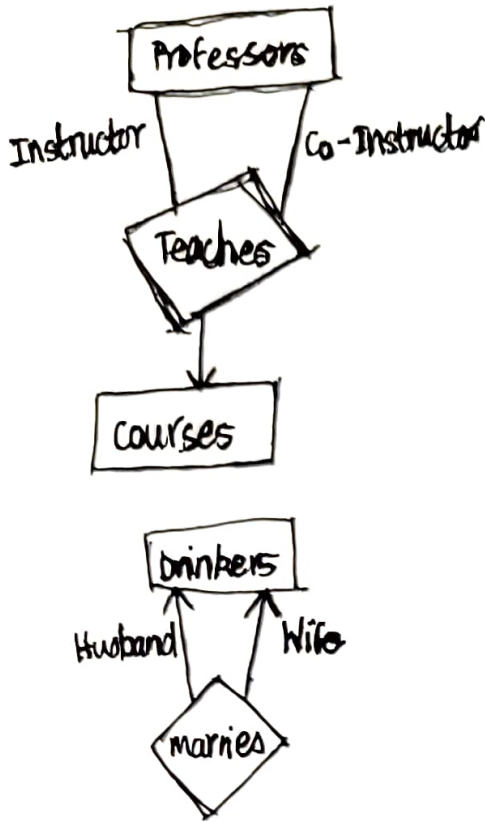
"Exactly-One" in a Relationship

- Use Round-Arrow



Roles in Relationships

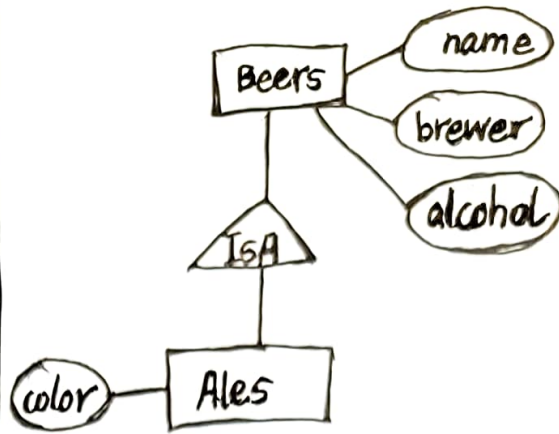
- When an entity set is involved multiple times in a relationship.
- Label each involvement with a role.



1.5 ER Model - subclasses

- Natural to think of real-world entities as specialization of others.
 - Specialization implies additional attributes.
 - Specialization implies fewer members.
- An ISA triangle indicates a subclass relationship.
- Directed - from subclass pointing to superclass.
- Subclass inherits all attributes of superclass and some more. (subclasses)
- No multiple inheritance (only one superclass). Subclasses form a tree.

3



1.6 ER Model - constraints

- Constraint is an assertion about the database that must be true at all times.
- Part of the data modeling (and database "schema")
- Important in database design.

Types of Constraints

- Keys: Unique Identifier
 - Ex. netid
- Referential Integrity: A referenced entity must exist.
 - If a drinker frequents a bar, the bar must exist.
- Many other constraints
 - Domain constraints: $Grade \in \{A, B, C, D\}$.
 $Price \in [0, 500]$
 - General constraints: $BirthDate < Employment Date$

Why are constraints important?

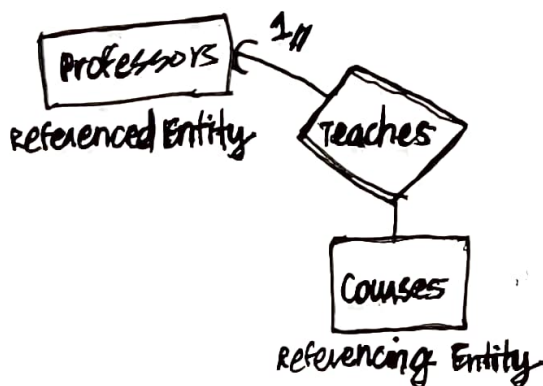
- Give more semantics to data.
- So we can check correctness (consistency) of data.
- So we can enable efficient storage, data lookup, etc.

Keys

- A key can consist of one or more attributes.
 - {id} identifies a student.
 - {name, major} identifies a student.
- Every entity set must have a key.
 - At least the entire set of attributes would form a key.
 - We must capture sufficient attributes to distinguish entities.
- There may be more than one key.
 - We designate one key as the primary key by underlining its attributes.
 - No formal way to specify multiple keys other than primary.

Referential Integrity Constraint

- In a relationship, expressed by a round arrow:
 - A "referencing" entity must reference exactly-one "referenced" entity.
 - A referenced entity must exist.
- I.e. no dangling pointer (segmentation fault in programming)



1.7 Beyond ER Model - UML

(4)

Unified Modelling Language

Other models exist but have similar concepts.

ER model is simple and natural.

ER model corresponds well to relational model.

- Relational model is a physical data model, used by relational databases.
- ER model is thus:
 - becomes popular conceptual design before concretizing to relational model.
 - provides semantic interpretation for physical "relations".
- ER model is graphical -- intuitive to create and view.
- Concepts of other models are similar.

UML

- Rectangular boxes for entity sets.
Entity set $\rightarrow$ class
- PK written for primary keys in UML
- Relationship $\rightarrow$ Association and Association classes.

UML allows binary associations only.

K-way can be expressed as K Binaries.

- A multiway relationship is just a combination of (zero or) one entity from each entity set.
- We can represent that combination as an entity.
- That new entity will connect to each entity set with arrows.

ER Relationship vs UML Association

	ER Model	UML
cardinality	binary, multiway	binary
Multiplicity	arrow, roundarrow, $\leq n$	m...n